REVIEW

Epidemiology and genetics of Meniere's disease

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Purpose of review

This review discusses the recent developments on the understanding of epidemiology and genetics of Meniere's disease.

Recent findings

Meniere's disease has been shown to be associated with several comorbidities, such as migraine, anxiety, allergy and immune disorders. Recent studies have investigated the relationship between environmental factors and Meniere's disease such as air pollution, allergy, asthma, osteoporosis or atmospheric pressure, reporting specific comorbidities in East Asian population. The application of exome sequencing has enabled the identification of genes sharing rare missense variants in multiple families with Meniere's disease, including *OTOG* and *TECTA* and suggesting digenic inheritance in *MYO7A*. Moreover, knockdown of *DTNA* gene orthologue in Drosophila resulted in defective proprioception and auditory function. *DTNA* and *FAM136A* knockout mice have been studied as potential mouse models for Meniere's disease.

Summary

While it has attracted emerging attention in recent years, the study of Meniere's disease genetics is still at its early stage. More geographically and ethnically based human genome studies, and the development of cellular and animal models of Meniere's disease may help shed light on the molecular mechanisms of Meniere's disease and provide the potential for gene-specific therapies.

Keywords

animal models, epidemiology, genetics, Meniere's disease

INTRODUCTION

Meniere's disease is a chronic disorder of the inner ear characterized by recurrent episodes of vertigo, tinnitus, sensorineural hearing loss (SNHL) and aural fullness. The cause and molecular pathogenesis of Meniere's disease is not fully understood. There is no biological marker for Meniere's disease diagnosis, and its diagnosis is based on the clinical registry of auditory and vestibular symptoms [1]. Endolymphatic hydrops, which is caused by the accumulation of endolymph resulting in enlargement of the endolymphatic space in the inner ear, has been one of the etiologic factors of Meniere's disease. However, endolymphatic hydrops itself does not explain the episodic nature of symptoms such as vertigo attacks or sudden tinnitus changes, since histopathological studies have reported endolymphatic hydrops in individuals without Meniere's disease symptoms [2]. The role of genetic and environmental factors in Meniere's disease pathogenesis has been extensively studied in the recent years.

Cluster analysis has identified five clinical variants or subtypes in Meniere's disease: classic/sporadic Meniere's disease without migraine; delayed Meniere's disease; familial Meniere's disease (FMD); migraine-associated Meniere's disease; and autoimmune-associated Meniere's disease [3]. These clinical groups are likely the results of different causes for the condition. In this review, we focus on the recent developments and understanding of Meniere's disease epidemiology and genetics.

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KEY POINTS

- Meniere disease is a multifactorial disorder with a relevant genetic contribution, and it is likely that different causes are responsible for different clinical subtypes.
- Meniere disease is associated with migraine, immune-related disorders and allergy. Some environmental factors such as air pollution, seasonal changes or allergens may also contribute to trigger Meniere crises.
- Familial Meniere's disease has a complex heritability, including autosomal dominant, autosomal recessive and multiallelic inheritance, the most common genes being *OTOG, MYO7A* and *TECTA*.
- Variants of genes involved in hearing loss, axon guidance, oxidative stress and autoimmune disorders have been reported in sporadic Meniere disease patients.
- Animal models of Meniere's disease are needed to understand the molecular mechanisms behind this condition.

EPIDEMIOLOGY

The prevalence and incidence of Meniere's disease varies according to the ethnic and geographic background in the world, ranging from 3 to 513 per 100 000 individuals. Meniere's disease is more frequent in European descendant populations, with a North to South gradient in Western Europe. The highest prevalence of 513 cases per 100 000 was found in Finland, where many families were also reported, and the lowest was found in Cantabria, Spain, with 75 cases per 100 000. In the United States, the prevalence was found to be 190 per 100 000. People of Asian or African ancestry have lower incidence rate, with 3.5 cases per 100 000 inhabitants in Japan [3,4]. A prevalence of 70.4 cases per 100 000 in Taiwan was noted in a nationwide study [5].

The condition has an average age of onset of 30–50 years, and it seldom affects children. Onset in older people is rare, with just 10% of patients showing disease onset after 65 years. Its prevalence increases with age. In the United States, Meniere's disease prevalence varied from 9 cases per 100 000 in patients under 18 years to 440 cases per 100 000 in patients over 65 years [4]. An epidemiological study in Taiwan showed peak age onset of Meniere's disease was shifted from 45–54 years during 2001–2010, to 55–64 years during 2011–2020. Likewise, mean ages of Meniere's disease also increased. The incidence of Meniere's disease patients over 65 years also increased from 12.4% (2001–2010) to 18.2% (2011–2020), which is consistent with previous reports [6].

Most studies have reported a female preponderance of Meniere's disease, the greatest differences in sexes being described in northern Europe, with a women ratio of 4.3 : 1 in Finland [3]. However, some more recent studies have noted similar frequencies among men and women. A study in United States suggests that Meniere's disease susceptibility is highest in older populations, white race, from higher income and rural areas, with a female to male ratio of only 1.25 [7]. The population-based study in Taiwan also showed higher prevalence in rural areas, with a female to male ratio of 1.84 [5].

MD can have a unilateral or bilateral involvement. Although unilateral Meniere's disease (UMD) is more frequent, bilateral Meniere's disease (BMD) increases with duration of illness, and up to 45% of unilateral Meniere's disease patients will develop contralateral involvement over time. The age of Meniere's disease onset, high-frequency hearing loss in the first audiogram and migraine may help to assess the risk to develop bilateral SNHL in Meniere's disease, facilitating the management of the condition [8]. In an Asian Meniere's disease patient study, the prevalence of BMD was 15.4%. The disease duration of BMD was significantly longer than that of UMD. Higher frequencies of delayed Meniere's disease and family history of vertigo were found in patients with BMD compared with patients with UMD. Endolymphatic hydrops was observed in 100% of cases on the clinically affected side and 6.1% of cases on the unaffected side [9].

Meniere's disease is a multifactorial disorder and several associated conditions have been considered as triggers or comorbidities (Fig. 1). Previous studies suggested that the prevalence of anxiety and depression is higher in Meniere's disease than healthy individuals [10]. However, the most consistent associations are with migraine and immune-related disorders [11▪▪]. A population-based study in Korean also supports that Meniere's disease and migraine demonstrated bidirectional associations. Patients with Meniere's disease had a higher risk of developing migraine, and patients with migraine had an increased risk of Meniere's disease [12]. Another large study investigated the comorbidities associated with Meniere's disease in Asian population and found an increase in the incidence of allergic rhinitis and allergic asthma. However, this study failed to find significant differences in the incidence of autoimmune diseases, which is inconsistent with previous studies [13]. A history of asthma was also reported to be associated with slightly higher incidence of Meniere's disease [14].

The role of vitamin D and osteoporosis has also been investigated in Meniere's disease. A recent study also demonstrated bidirectional associations

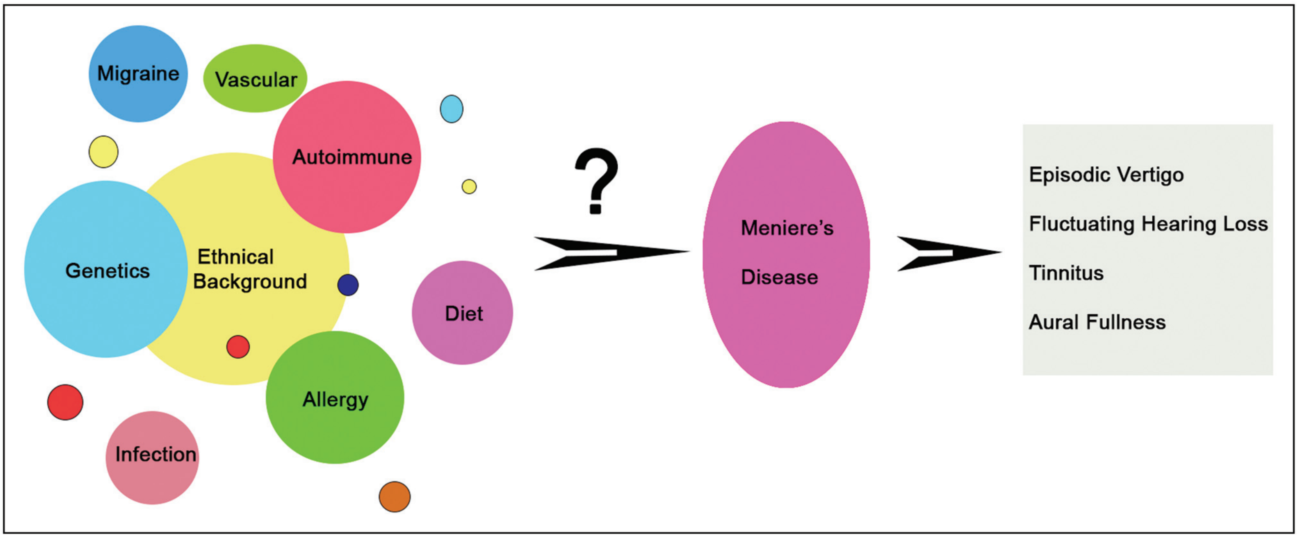

FIGURE 1. Schematic representation of factors contributing to Meniere disease development.

between Meniere's disease and osteoporosis. Adult patients with Meniere's disease had a greater risk of osteoporosis. In addition, adult patients with osteoporosis also showed a higher risk of Meniere's disease [15]. Significantly lower serum vitamin D level was noted in another study [16]. Air pollution was also shown to be associated with higher hospital visits for Meniere's disease [17]. One Iranian study measured the immunoreactivity to food allergens and found that Meniere's disease patients are more reactive than control group [18]. One study using peripheral blood mononuclear cell (PBMC) from Meniere's disease patients showed a similar reaction with other autoimmune disease to NaCl exposure, with increased levels of IL1β [19].

The global COVID-19 pandemic has greatly affected people's mental health. A preliminary Italian study showed higher incidence of Meniere's disease in 2020 than previous years, which indicated stress may be the leading cause [20]. Conversely, a study in Taiwan found lower incidence of Meniere's disease in 2020 compared to the previous 4 years [21]. This may be explained by mask wearing to avoid temperature change since Meniere attack may be associated with atmospheric pressure and seasonal changes [22]. Further studies are needed to gain a better understanding on the impact of COVID-19 pandemic in Meniere's disease.

GENES AND FAMILIAL MENIERE'S DISEASE

The vast differences of Meniere's disease prevalence in different ethnic groups and the demonstration of familial clustering in Europeans support a genetic contribution to Meniere's disease cause [23]. Moreover, FMD has been repeatedly described in European descendent population in 5–20% of cases [2], but these numbers may underestimate the genetic contribution, since Meniere's disease relatives with isolated SNHL or episodic vertigo could be incomplete phenotypes (partial syndromes) and they are not diagnosed as Meniere's disease.

Familial Meniere's disease is a heterogeneous condition and multiple genes have been reported, mostly in European descendant population [24]. The first exome sequencing study in FMD was published in 2015 in a Spanish family with three women in consecutive generation with Meniere's disease that segregated a nonsense variant in the *FAM136A* gene and a missense variant in *DTNA* gene.

More than 100 families with Meniere's disease have been sequenced and the most relevant gene is *OTOG*, which encodes for Otogelin (Uniprot Q6ZR10), a structural protein involved in the organization of the tectorial and otolithic membranes, which play a role in the anchorage of the stereocilia tip to the tectorial membrane [25]. Rare missense variants segregate in 6 families and support a compound recessive inheritance [26]. Rare variants in the genes *OTOG*, *MYO7A*, *CDH23*, *ADGRV1* and *TECTA* have been reported in multiplex Meniere's disease families with autosomal dominant, autosomal recessive and digenic inheritance (Table 1) [27▪].

Other candidate genes identified in FMD include *FAM136A*, *DTNA*, *PRKCB*, *COCH*, *DPT*, *SEMA3D*, *GUSB*, *SLC6A7*, *STRC*, *HMX2*, *TMEM55* and *LSAMP* [26,28]. The limitation of these segregation studies is that those variants were only found in one family, although *DTNA, FAM136A, DPT* mutations have been also found in sporadic Meniere's disease

Table 1. Genes and variants reported in multiplex Meniere disease families

					Allelic frequency[c]			
Gene	**Chr**	**Position[b]**	**ID**	**Protein**	**gnomAD**	**Other**	**Inheritance pattern**	**CADD score**
TECTA	11	121152980	rs774697277	p.Cys1402Ser	3.33×10^{-5}	10^{-3} (CSVS) 3.3×10^{-5} (ExAC)	AD	28.00
		121157956	-	p.Asn1474LysfsTer91	Novel			-
		121158016	rs200544452	p.Val1494Ala	6.57×10^{-5}	10^{-3} (CSVS) 5.9×10^{-5} (ExAC)		24.30
		121165368	-	p.Pro1790Ser	Novel			16.10
		121189864	rs1223512271	p.Gly2118ProfsTer22	6.58×10^{-6}	NF (CSVS)		35.00
OTOG	11	17553211	rs552304627	p.Val141Met	8.35×10^{-4}	4×10^{-3} (CSVS) 4.1×10^{-4} (ExAC)	AR	33.00
		17557227	rs61978648	p.Val269Ile	2.04×10^{-2}	1.4×10^{-2} (CSVS) 8.01×10^{-3} (ExAC)		19.12
		17573200	-	p.Pro747Thr	Novel			21.90
		17599671	rs117005078	p.Pro1240Leu	3.3×10^{-3}	4×10^{-3} (CSVS) 1.68×10^{-3} (ExAC)		33.00
		17606001	rs145689709	p.Arg1353Gln	2.84×10^{-3}	6×10^{-3} (CSVS) 1.98×10^{-3} (ExAC)		22.00
		17609906	rs117380920	p.Leu1548Phe	8×10^{-3}	1.3×10^{-2} (CSVS) 1.07×10^{-2} (ExAC)		12.42
		17611374	rs61736002	p.Ala2037Val	2.41×10^{-3}	4×10^{-3} (CSVS) 3.74×10^{-3} (ExAC)		7.61
		17635125	rs76461792	p.Arg2556Gln	3.06×10^{-3}	4×10^{-3} (CSVS) 2.95×10^{-3} (ExAC)		23.50
		17642200	rs117315845	p.Arg2802His	2.04×10^{-3}	6×10^{-3} (CSVS) 3.68×10^{-3} (ExAC)		16.79
		17645592	rs61997203	p.Lys2842Asn	1.57×10^{-2}	1.9×10^{-2} (CSVS) 9.79×10^{-3} (ExAC)		24.20
MYO7A	11	77130637	rs782787126	p.MetIc[d]	2.02×10^{-5}	NF (CSVS) 9×10^{-6} (ExAC)	Digenic	24.00
		77159450	rs45629132	p.Arg336His	1.61×10^{-3}	4.9×10^{-4} (CSVS) 1.15×10^{-3} (ExAC)		24.10
		77174877	rs781991817	p.Arg686His	3.54×10^{-4}	3.7×10^{-3} (CSVS) 1.6×10^{-4} (ExAC)		29.80
		77179874	rs782179888	p.Arg836His	4.6×10^{-5}	NF (CSVS) 2.3×10^{-4} (ExAC)		24.50
		77180404	rs200454015	p.Arg873Trp	8.41×10^{-4}	2.4×10^{-4} (CSVS)		24.80
		77199601	-	p.Trp1545[a]	Novel			43.00
		77211830	rs41298759	p.Ala2083Thr	2.63×10^{-4}	2.4×10^{-4} (CSVS) 4.16×10^{-4} (ExAC)		21.80
		77214674	rs776881443	p.Arg2209Gln	1.97×10^{-5}	NF (CSVS)		22.80
		77214688	rs111033231	p.Gly2214Ser	1.74×10^{-2}	10^{-3} (CSVS) 8.85×10^{-3} (ExAC)		10.40
ADGRV1	5	90694338	rs201733037	p.Pro2528Ser	3.96×10^{-3}	5×10^{-3} (CSVS) 4.3×10^{-3} (ExAC)	Digenic	22.30
		90840606	rs200907244	p.Arg5547His	1.91×10^{-4}	2.5×10^{-4} (CSVS) 1.02×10^{-4} (ExAC)		19.50
CDH23	10	71732116	rs149073355	p.Asn1282Ser	3.33×10^{-3}	4×10^{-3} (CSVS) 2.82×10^{-3} (ExAC)	Digenic	23.20
		71793440	rs531513127	p.Arg2176His	6.57×10^{-5}	2.5×10^{-4} (CSVS)		20.20

CADD, Combined Annotation Dependent Depletion; CSVS, Collaborative Spanish Variant Server; ExAC, Exome Aggregation Consortium; gnomAD, Genome Aggregation Database, ID reference Single Nucleotide Polymorphism identifier, NF not found.
[a]Stop codon.
[b]Positions have been updated according to the GRCh38/hg38 reference genome.
[c]Allelic frequencies reported in the original reports have been updated according to the available information in the last version of the reference database (gnomAD v3.1.2).
[d]Start loss.

patients in East Asian population, functional studies in cellular or animal model are lacking [29].

ANIMAL MODELS IN FAMILIAL MENIERE'S DISEASE

Three animal models have been generated to perform functional studies in FMD. In a Drosophila model, knockdown of the *DTNA* orthologue in Drosophila, Dystrobrevin (Dyb), resulted in defective proprioception and impaired function of Johnston's Organ, the fly's equivalent of the inner ear [30■■]. Further study explored the function of *FAM136A* and *DTNA* using two transgenic mouse models. The *FAM136A* knockout mice showed a hearing loss compared to their wild-type counterparts. While *DTNA* knockout mice showed a significantly decreased loss of balance. Both mouse models exhibit an expected age-related loss of balance and a learning effect due to repeated trials during training [31■].

GENES AND SPORADIC MENIERE'S DISEASE

There are two studies in sporadic that used gene panels in a large cohort of patients with SMD. These studies found an enrichment of rare coding variants in some hearing loss genes and axonal-guidance signalling pathways (Table 2) [32,33]. Those SNHL genes include *GJB2*, *USH1G*, *SLC26A4*, *ESRRB*, *CLDN14* and *MARVELD2* [32]. Moreover, *NTN4* gene is associated with axon guidance signalling pathway. A significant burden of damaging variants in the *NTN4* gene was found in SMD cases, suggesting that axon guidance pathway may contribute to the genetic architecture of SMD [33]. However, those studies were limited to Spanish and Portuguese population.

A recent preliminary study in a small cohort of Chinese patients identified several variants that involved in autoimmunity and autoinflammation [34]. A larger sized study is needed to validate the findings.

EPIGENETICS AND MENIERE'S DISEASE

A pilot epigenomic study was performed using whole-genome bisulfite sequencing (WGBS) suggesting that the DNA methylation signature could allow distinguishing between Meniere's disease patients and controls [35]. In this study, a great number of differentially methylated CpGs were found when comparing Meniere's disease patients to controls. Of note, few of these CpGs involved several hearing loss genes, including *CDH23*, *PCDH15* or *ADGRV1*, that encode for stereocilia link proteins.

Another study measured microRNA expression in Meniere's disease patients. They found that

Table 2. Genes and rare variants reported in sporadic Meniere's disease in European population

Gene	Chr	Position[a]	ID	Protein	Allelic frequency (gnomAD v3)[b]	CADD score
ESRRB	14	76491548	rs201344770	p.Asp318Asn	2.56×10^{-4}	30.00
		76499993	rs200237229	p.Arg476His	5.52×10^{-4}	2.57
		76500004	rs201448899	p.Pro480Ser	8.80×10^{-4}	11.30
MARVELD2	5	69419994	rs369265136	p.Leu203Leu	6.57×10^{-6}	8.79
SLC26A4	7	107695963	rs200511789	pIle490Leu	1.77×10^{-4}	22.2
USH1G	17	74919824	rs151242039	p.Gly338Arg	6.63×10^{-4}	14.9
		74920448	rs111033465	p.Lys130Glu	3.62×10^{-2}	24.6
GJB2	13	20189125	rs111033186	p.Val153Ile	$4.83x10^{-2}$	12.3
		20189473	rs72474224	p.Val37Ile	4.00×10^{-3}	21.7
		20189313	rs80338945	p.Leu90Pro	$6.37x10^{-4}$	29.7
		20189503	rs2274084	p.Val27Ile	$2.59x10^{-2}$	23.3
NTN4	12	95787225	rs537136429	p.Arg100Gln	$7.2x10^{-5}$	17.4
		95787307	rs34114770	p.Arg73Trp	$3.69x10^{-3}$	24.9
		95787424	rs199956955	p.Cys34Arg	$1.12x10^{-4}$	28.4

CADD, Combined Annotation Dependent Depletion; gnomAD, Genome Aggregation Database, ID reference Single Nucleotide Polymorphism identifier.
[a]Positions have been updated according to the GRCh38/hg38 reference genome.
[b]Allelic frequencies reported in the original reports have been updated according to the available information in the last version of the reference database (gnomAD v3.1.2).

Meniere's disease patients exhibit distinct micro-RNA expression profiles within both the perilymph and serum compared to controls [36]. However, these studies were performed in small series of sporadic patients with Meniere's disease and further studies are needed to clarify the role of epigenetics in Meniere's disease.

CONCLUSION

Epidemiological studies have been carried out in Asian population in the recent years. However, the complexity of Meniere's disease epidemiology is partly due to the challenge of clinical diagnosis, changes in diagnostic criteria over time, the different methodologies used and the populations surveyed. Our understanding of Meniere's disease genetics is still limited, but familial Meniere's disease studies support autosomal dominant and recessive inheritance, *OTOG*, *MYO7A* and *TECTA* being the most commonly found genes. Human genome sequencing studies in large Meniere's disease cohorts are needed. Genetically modified animal models are essential tools to understand the molecular mechanisms of Meniere's disease pathogenesis.

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Conflicts of interest

There are no conflicts of interest.

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■■ of outstanding interest

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